NANYANG TECHNOLOGICAL UNIVERSITY

SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2 MH1301 Discrete Mathematics Tutorial 10

For the tutorial in Week 11 on 4 April, let us discuss the following questions.

Question 1. (Ex 10.7.8, 10.7.13.)

(a) Suppose that a connected planar graph has eight vertices, each of degree three. Into how many regions is the plane divided by a planar representation of this graph ?

(b) Suppose that a connected planar simple graph with m edges and n vertices contains no simple circuits of length 4 or less. Show that $m \leq (5/3)n - (10/3)$ if $n \geq 4$.

Question 2.

(a) Prove that $K_{2,5}$ is planar by drawing the graph without edges that cross each other. What about $K_{2,n}$?

(b) Prove that there are no 6-regular simple graphs that are planar. (Hint: How many edges does a 6-regular graph have?)

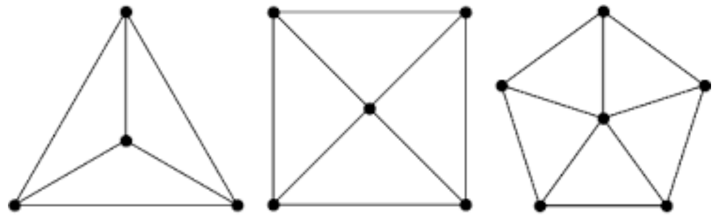
Question 3. Show that if G is a simple graph with at least 11 vertices, then either G or $\overline{G}$, the complement of G, is nonplanar.

Question 4.

(a) Find a planar 3-regular graph.

(b) Is there a planar graph with 10 vertices and 25 edges?

Question 5. Determine the chromatic numbers of the following three graphs (called *wheel graphs*).



$\mathbf{W_4}$ $\mathbf{W_5}$ $\mathbf{W_6}$

Question 6.

(a) Under what condition on n is there a graph with n vertices where all vertices have degree 1?

(b) True or false: a planar graph that has no subgraph that is an odd cycle can be 2-colored.

Question 7.

(a) How many edges does one have to add to the wheel graph W_5 (see above) so that the resulting (simple) graph is non-planar?

(b) If we add three new edges to W_6 (so that the result is a simple graph), can the result be planar?

Question 8.

(a) Is there a connected planar graph with n vertices and $2n$ regions?

(b) Is there a connected planar graph with n vertices and n regions?

Question 9.

(a) Let $n \geq 3$. Is there a connected, planar, bipartite graph with n regions and n vertices?

(b) Is there a connected planar graph with 6 vertices and 8 regions?

Answer.

1(a) 6.
2(a) planar.
5) 4, 3, 4.
6(b) True.
7) 2; No.
8) No; Yes.
8) No; Yes.